

# Mahmoud A. Aboulata

## Personal Information

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| Address | New Cairo, Cairo - Egypt                                                     |
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| Email   | <a href="mailto:mahmoudaboulata01@gmail.com">mahmoudaboulata01@gmail.com</a> |
| Links   | <a href="#">LinkedIn</a>   <a href="#">GitHub</a>                            |

## Interests and Goals

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I have a mix of project experience in Data Engineering, Machine Learning, and pipeline systems design. My interests are to apply end-to-end data infrastructure and automated analytics stacks to solve complex, real-world problems.

## Skills & Tech Stack

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| ○ Programming & Query Languages     | - Python · SQL · Bash · Java · C++ · html/Js/Css                                                                                                                                                          |
| ○ Data Engineering & Infrastructure | - Apache Kafka · Apache Spark · Apache Airflow · dbt<br>Debezium · Delta Lake · PostgreSQL · Docker · Kubernetes · Terraform · AWS                                                                        |
| ○ ML/AI & Data Science              | - Scikit-learn · XGBoost · TensorFlow · PyTorch · HuggingFace<br>SHAP · MLflow · Evidently AI · FastAPI<br>- Pandas · NumPy · Matplotlib · Seaborn · Plotly<br>Statsmodels · Prophet · Power BI · Tableau |
| ○ Advanced Methods                  | - Feature Engineering · Time Series Analysis · A/B Testing<br>Causal Inference · Survival Analysis · NLP · Anomaly Detection                                                                              |
| ○ Data Governance & Compliance      | - PII Handling · Model Auditing · Fairness Analysis<br>Egyptian PDPL · GDPR Article 22 · Audit Logging                                                                                                    |
| ○ Cloud & DevOps                    | - AWS (S3 · Glue · Redshift · Kinesis · Lambda)<br>GitHub Actions · CI/CD · Linux<br>AWS Certified Data Engineer Associate                                                                                |

## Education

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|------|------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | - BSc. (Hons) Informatics & Computer science<br>(Artificial Intelligence Major)<br><br><b>GPA:</b> 3.1/4<br><b>Graduation Year:</b> 2028 |
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## Experience & portfolio - [GitHub](#)

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### Independent (Freelance) & Personal projects:

All projects are either through independent contracts, freelancing or personal exercising | **Started 2026**

#### Automated Data Platform

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|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| • Project                  | - <b>Egyptian Banking &amp; Financial Inclusion Analysis</b>                                                                                             |
| • Ingestion & Processing   | - Built an automated Python and API data extraction pipeline from scratch with input validation and exception handling.                                  |
| • Orchestration & Database | - Managed processing dependencies using Apache Airflow, transforming raw data into optimized PostgreSQL relational schemas via CTEs and indexed queries. |

#### Real-Time Streaming & Distributed Infrastructure

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|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| • Project         | - <b>E-Commerce Financial Analytics Pipeline</b>                                                                                                |
| • Event Streaming | - Engineered an end-to-end Kafka event streaming platform using PySpark Structured Streaming with watermarking to manage late-arriving metrics. |
| • Analytics Layer | - Implemented a containerized, multi-tier dbt model architecture from raw staging to final presentation marts with built-in data tests.         |

#### End-to-End Predictive ML System

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|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| • Project                   | - <b>Credit Default Prediction with Explainability</b>                                                                                                 |
| • Management & Optimization | - Built complete batch ingestion and ML system, tracking parallel modeling algorithms and hyperparameter tuning via MLflow.                            |
| • Deployment & Scaling      | - Containerized the pipeline using Docker to minimize image sizes, deploying a high-availability API on Kubernetes with zero-downtime rolling updates. |

#### GOVERNANCE, RISK MANAGEMENT & COMPLIANCE

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|--------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| • Projects         | - <b>Real-Time Financial Transaction Monitoring Platform</b>                                                                        |
|                    | - <b>3D Printing Operations Intelligence Platform</b>                                                                               |
| • Data Privacy     | - Programmed algorithmic PII masking/anonymization loops compliant with Egyptian PDPL (Law 151/2020) and GDPR.                      |
| • Model Validation | - Audited algorithmic bias and prevented data leakage using temporal validation splits, SHAP values, and demographic parity checks. |

